Deep Learning

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Course Learning Objectives:

- To understand the fundamentals of deep learning.
- To know the main techniques in deep learning and the main research in this field.
- Be able to design and implement deep neural network systems.
- Be able to identify new application requirements in the field of computer vision.
- Be able to identify reasonable work goals and estimate the resources required to achieve the objectives.
- Be able to structure and prepare scientific and technical documentation describing project activities.
- Be able to autonomously extend the knowledge acquired during the study course by reading and understanding scientific and technical documentation.

Course Content

UNIT-I **(08 HOURS)**

Foundations of Deep Learning

- Overview of deep learning.
- Importance of advanced deep learning in contemporary applications
- Introduction to neural networks
- How does a neural network work
- Key concepts of neural networks, layers, and activations
- Training neural networks with backpropagation
- Optimization techniques

UNIT-II **(08 HOURS)**

Convolutional Neural Networks (CNNs)

- Overview of CNN architecture
- Convolution operation in CNN
- Padding in convolutional neural networks
- Convolutional layers and filters
- Max pooling layer in CNN
- Data augmentation in CNN
- Addressing bias and fairness in CNN models

UNIT-III **(08 HOURS)**

Recurrent Neural Networks (RNNs) and LSTMs

- Overview of recurrent neural networks
- Applications of RNN
- Recurrent neural network forward propagation
- Backpropagation in RNN
- Vanishing and exploding gradient problems
- Working procedure of Long Short-Term Memory (LSTM) networks
- Overview of bidirectional LSTM

UNIT-IV **(08 HOURS)**

Building an End-to-End Deep Learning Project Pipeline

- Problem formulation and defining project objectives
- Data collection, cleaning, and preprocessing
- Exploratory data analysis and feature engineering
- Selecting and designing the model architecture
- Model training, hyperparameter tuning, and validation
- Model evaluation and performance metrics
- Model deployment strategies (APIs, cloud, and edge devices)
- Monitoring, logging, and maintaining models in production
- Case study: building a complete deep learning pipeline from raw data to a deployed application

Text Books

T1. Deep Learning - Goodfellow, Y. Bengio, and A. Courville, MIT Press, 2016
T2. Neural Networks and Deep Learning: A Textbook - C. C. Aggarwal, Springer, 2018
T3. Advanced Deep Learning with TensorFlow 2 and Keras - R. Atienza, Packt Publishing, 2020
T4. Deep Learning for Computer Vision - R. Shanmugamani, Packt Publishing, 2018

Reference Books:

R1. Deep Learning: A Practitioner's Approach - Adam Gibson and Josh Patterson, O'Reilly Media, 2017

Course Learning Outcomes (CLO)

CLO	Description	Bloom's Taxonomy Level
CLO1	To understand the fundamentals of deep learning.	1,1,2,3 Remembering, Understanding, Apply
CLO2	Be able to design and implement deep neural network systems.	3,4,5 Applying, Analyzing, Evaluating
CLO3	Be able to identify reasonable work goals and estimate the resources required to achieve the objectives.	3,4,5 Applying, Analyzing, Evaluating
CLO4	Be able to autonomously extend the knowledge acquired during the study course by reading and understanding scientific and technical documentation.	3,4,5 Applying, Analyzing, Evaluating

Mapping of CLOs with PLOs & PSOs

Course Learning Outcomes	Program Learning Outcomes (PLO's)														
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**	**PSO1**	**PSO2**	**PSO3**
CLO1	M	L	M	M			M		M				H	M	M
CLO2	M		M	M	M	M	H	H	M			H		M	
CLO3	H	H	H	H	M	M	H	H	M			H		M	
CLO4	H	H	H	H	M	M	H	H	M			H			M